

Clinical Classifications Software Refined (CCSR) for ICD-10-PCS

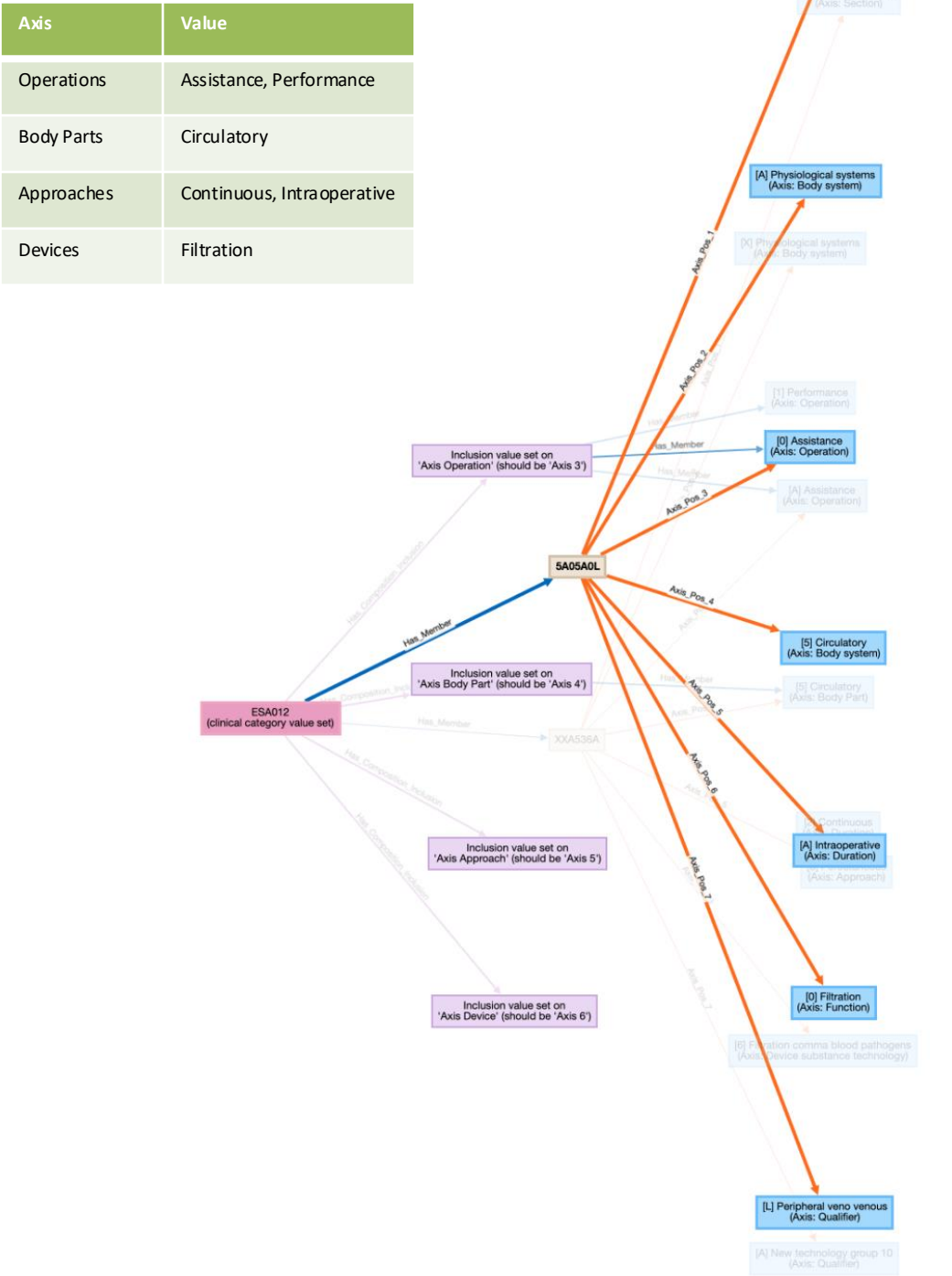
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Introduction & Methods

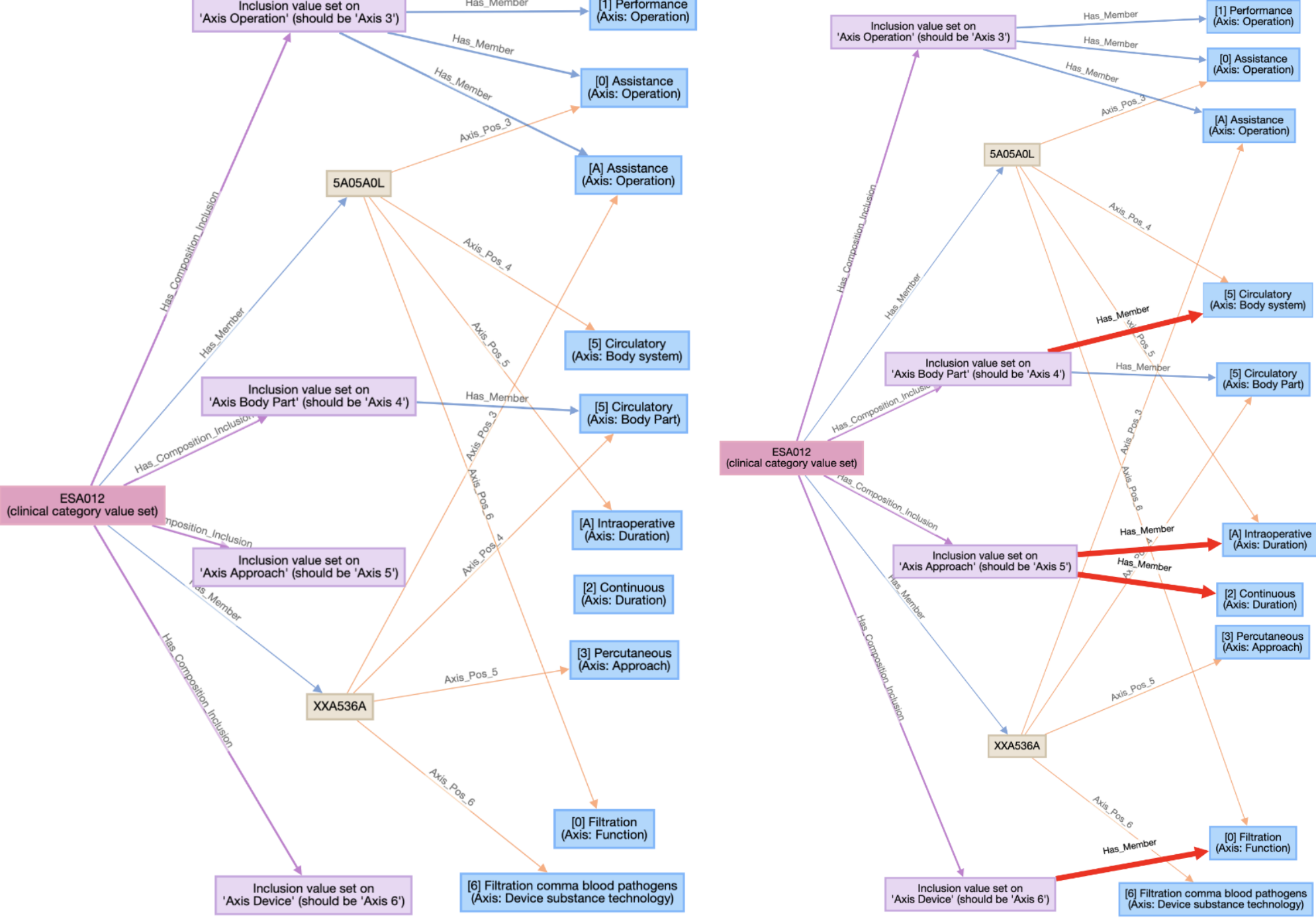
The CCSR value set's inclusion criteria were represented as **queries** on the knowledge graph reasoner – an **'AND' logic** is used between the 4 axes, and an **'OR' logic** is used for the multiple text descriptions for a given axis.

NODE DETAILS	
Extracorporeal Peripheral Veno-venous Filtration, Intraoperative	
Type:	Procedure
db_id	5A05A0L
description	Extracorporeal Peripheral Veno-venous Filtration, Intraoperative
display_label	5A05A0L
status	active
table	5A0
version	April-2025
Outgoing Relationships:	
BELONGS_TO	5A0 [Table]
Axis_Pos_1	[5] Extracorporeal or systemic assistance and performance (Axis: Section) [CharacterCode]
Axis_Pos_2	[A] Physiological systems (Axis: Body system) [CharacterCode]
Axis_Pos_3	[0] Assistance (Axis: Operation) [CharacterCode]
Axis_Pos_4	[5] Circulatory (Axis: Body system) [CharacterCode]
Axis_Pos_5	[A] Intraoperative (Axis: Duration) [CharacterCode]
Axis_Pos_6	[0] Filtration (Axis: Function) [CharacterCode]
Axis_Pos_7	[1] Peripheral veno venous (Axis: Qualifier) [CharacterCode]
Has_Member	Extracorporeal or systemic assistance and performance, NEC [ClinicalCategory]
LEGENDS	
ClinicalDomain	
ClinicalCategory	
ClinicalCategory2024	
Procedure	
CharacterCode	
InclusionCollection	

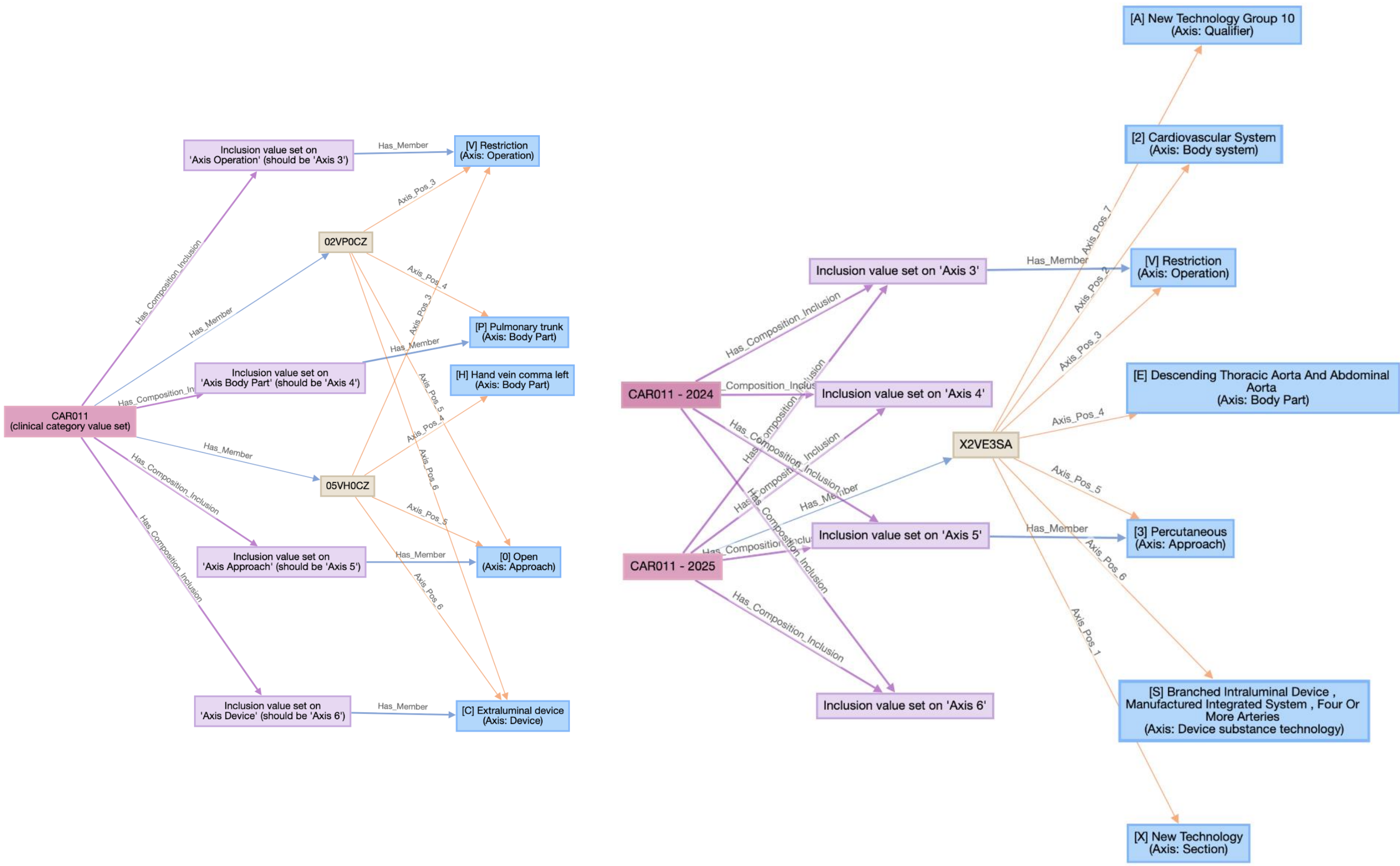


Results

For value set 'ESA012', specifying the 4 axis names as inclusion criteria in the query **yielded no results**. We replaced them with axes on positions 3 to 6, and **correctly yielded** the only **non-X-section concepts** '5A05A01'.



Results (Cont.)



1. **80.1%** concepts (N = 891) were **identified** in value set CAR011, with 93 'body part' in inclusion
 - Adding the **20 missing** 'body part', **16.2%** (N = 180) missing concepts (all **non-Section-X**) can be identified.
2. The **section-X concepts** (0.2%, N = 2) **not recoverable**
 - Neither in ESA011
 - Section X is 'New Technology'.
 - New rule exploration on Axis 1 and 2
3. **3.5%** (N = 39) retired concepts **have no linkage to construct the query and verify**
 - **Similar to without KG**

```
graph LR
    A[Value set] -- Has_Member --> B[Retired procedure]
```

Conclusions

- Several pitfalls when applying the 4-axis inclusion criteria for CCSR value sets using the knowledge graph granular representation:
 - (a) the positions are referred inaccurately as axe names,
 - (b) the values of text descriptions are missing, and
 - (c) the inclusion criteria are incomplete.
- While the intrinsic structure of ICD-10-PCS may not support the definition of complete inclusion criteria for CCSR value sets, the knowledge graph validates that no spurious concepts are identified outside of the two value sets. The identified pitfalls and the new rule exploration demonstrates **the critical role of KGs** in propositional reasoning.

References

[1] Centers for Medicare & Medicaid Services. ICD-10 | CMS. <https://www.cms.gov/medicare/coding-billing/icd-10-codes>. Published February 26, 2025. Accessed March 24, 2025.

[2] Agency for Healthcare Research and Quality. Healthcare Cost and Utilization Project (HCUP). Clinical Classifications Software Refined (CCSR) for ICD-10-PCS Procedures. <https://hcup-us.ahrq.gov/toolssoftware/ccsr/prccsr.jsp>. Published February 2025. Accessed March 25, 2025.